

is completely removed from the logic circuit. Thus, transistors in the circuit can be reduced to 14 by removing the B inverter and so lowering the power consumption.

Results and simulations

All the circuits have been simulated in HSPICE with 1V supply voltage using a 32nm model for low-power

applications. In each simulation average power, power dissipation, and transmission delay have been calculated. From these results the product delay has also been found out and compared.

Appropriate bit sequences are entered in the input to cover all possible transitions for the decoder. A 2-4 decoder has 2 inputs from which it can generate $2^2 = 4$ different binary combinations. The 2 to 4 decoder is simulated for 500ms.

The waveforms in Fig. 8 show our proposed circuits using CNTFET working well and we compare their results in the following figures and tables.

Fig. 9 shows comparison between the average power of MOSFET, CNTFET, and mixed logic decoder circuits. From the figure it can be seen that the average power consumption in CNT based decoders is very much less than in CMOS decoders, and the least consumption is in 14T CNTFET decoder.

Fig. 10 compares power dissipation amongst MOS, CNTFET, and mixed logic decoders. It can be seen the power dissipation in CNTFET is very less as compared to MOS circuits and the 14T CNTFET (mixed logic design) decoder has the least power dissipation.

Fig. 11 shows PDP comparison between 20T CMOS decoders and mixed logic 15T and 14T MOSFET and CNTFET decoders. From the results we infer that the CNTFET based decoders are better than the MOSFET based decoders as all the CNTFETs yield better PDP than their respective MOSFET decoder. Moreover, the PDP is lowest for the 14T CNTFET decoder amongst all. Thus, it is the most efficient decoder as compared to the others.

Simulation results for power dissipation, average power consumption,

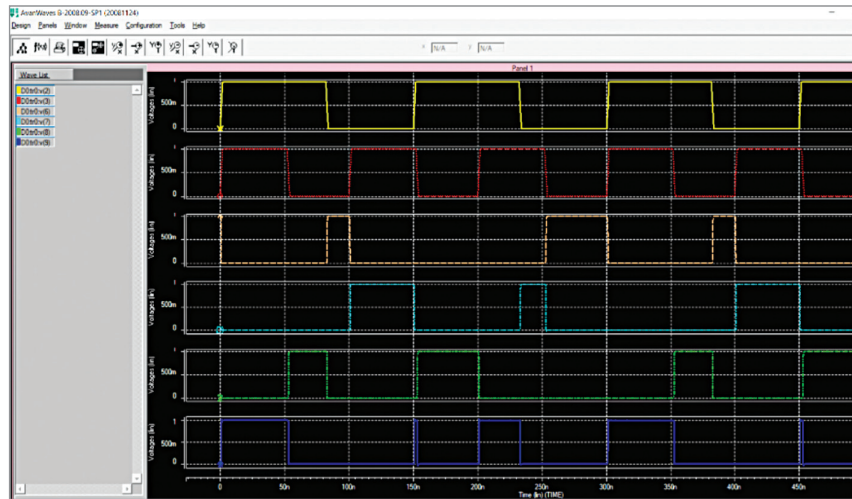


Fig. 8: Waveforms of 2 to 4 decoder

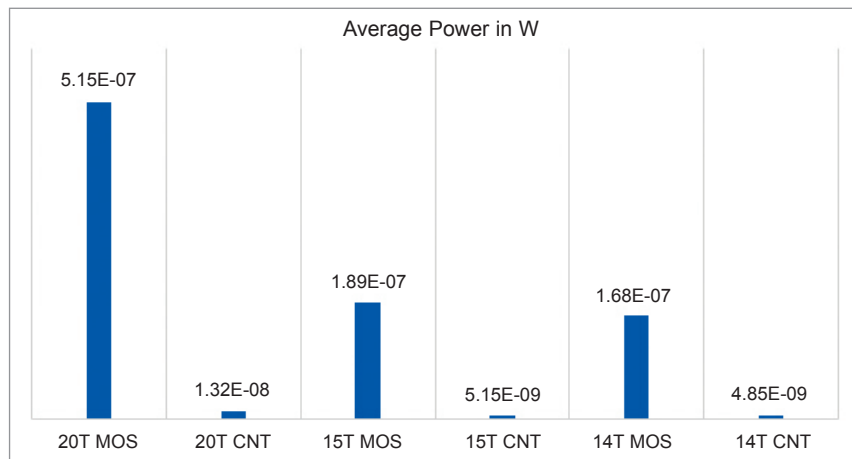


Fig. 9: Average power consumption comparison

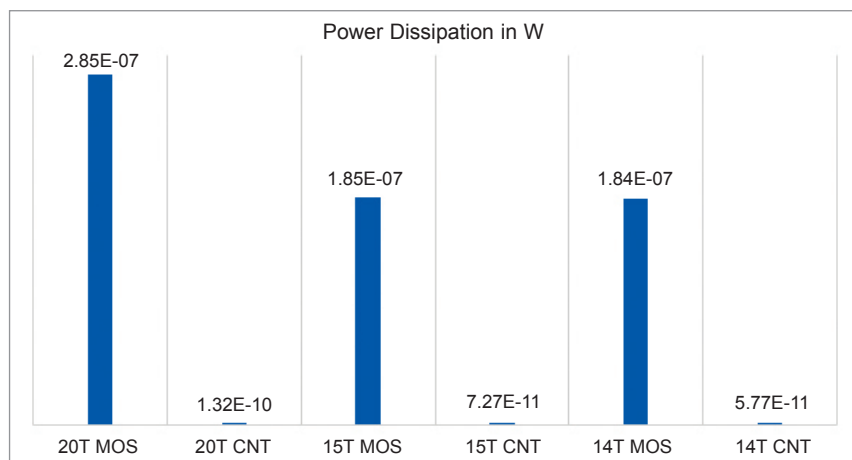


Fig. 10: Power dissipation comparison

20T CMOS v/s CNTFET Table 2		
Parameters	20T CMOS	20T CNTFET
Average Power in W	5.15E-07	1.32E-08
Delay in s	8.22E-08	8.21E-08
PDP in J	4.24E-14	1.08E-16
Power Dissipation in W	2.85E-07	1.32E-10